

Group members: Fatima Saldana- Gutierrez, Mariana Rivera, Sarah Boyd

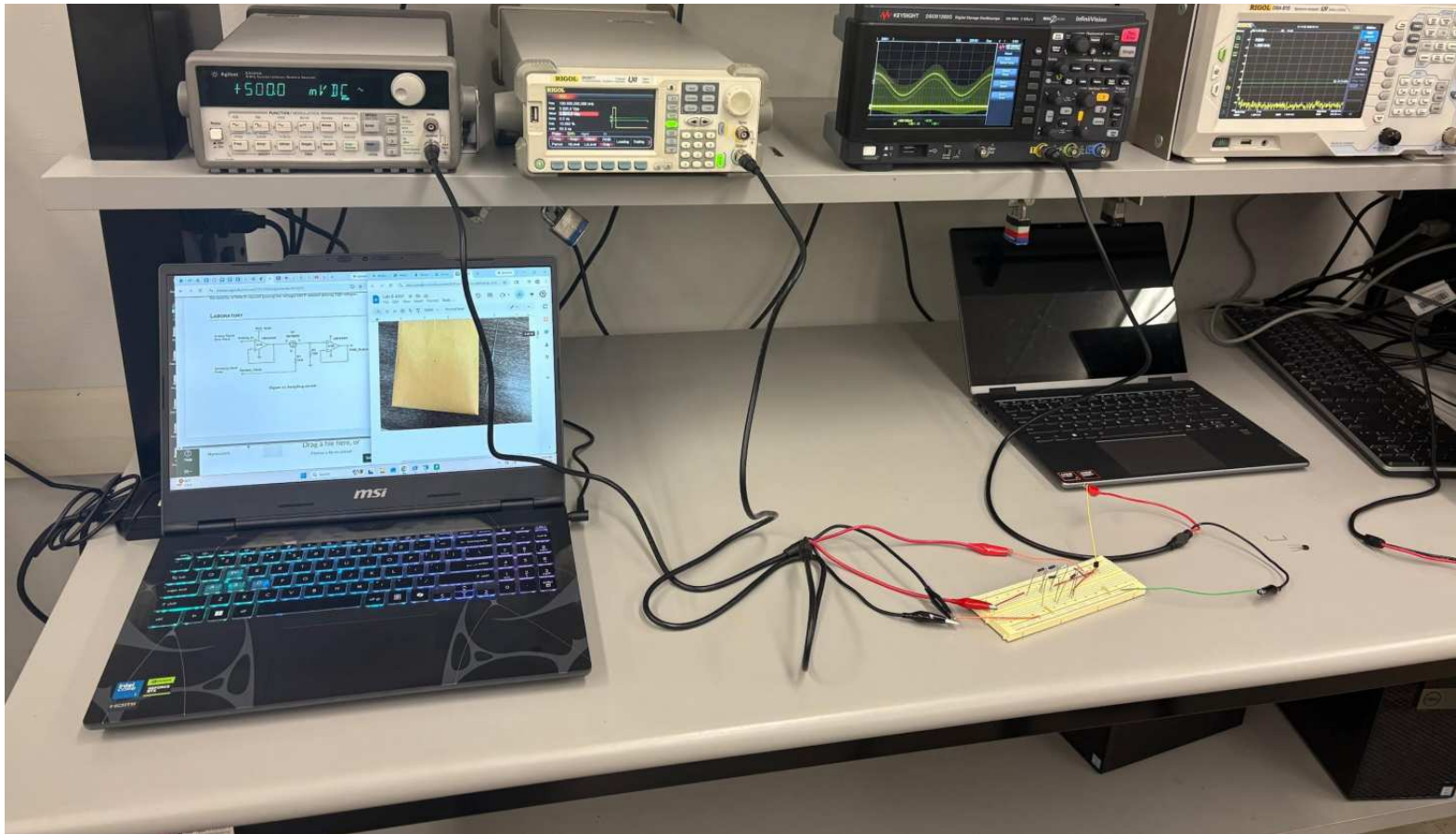
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ETE 4351 Spring 2026

Professor Jason Mandala

Lab 5

Sampling and Pulse Amplitude Modulation



OBJECTIVES

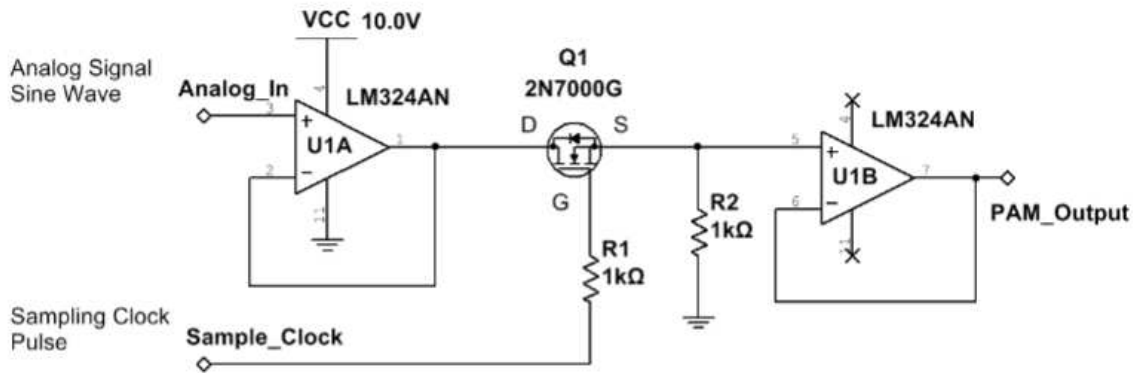
- Examine the effect of sampling a signal
- Understand the role of sampling frequency with respect to the Nyquist rate and aliasing
- Build and analyze a sampling circuit
- Reinforce use of laboratory equipment

Introduction

PAM (pulse amplitude modulation) is a modulation technique where an analog signal is represented by a series of pulses with different amplitudes. The amplitude of each pulse represents the amplitude of that analog signal at each time it was sampled. This process is familiar, as it is part of the process that an ADC (analog to digital converter) uses to make the data from the signal into different discrete levels. This process is called “sample and hold”. Sampling involves briefly turning a pass transistor or switch on at the sampling period T_s (which is a regular interval and is the inverse of the sampling frequency F_s), thus allowing the signal to momentarily pass through. ADC and its operation is essential to digital signal processing and digital control systems- the ADC converts an analog voltage into a digital binary value representing that voltage, allowing a processor to use the data (as processors only operate with digital data).

One of the most important things to consider when sampling any signal is the rate which the sampling occurs relative to the signal frequency. Recall that the Nyquist rate is used to find the lowest limit of the sampling frequency that is possible while still allowing for an accurate reconstruction of the signal. If the sampling frequency does not meet the Nyquist conditions, then an accurate reconstruction of the sampled signal is impossible and we would get an alias. According to the Nyquist theorem, we must sample a frequency at least twice the highest frequency present in the signal being sampled for accurate reconstruction of frequency ($F_s \geq 2f_{max}$) or ten times or greater for accurate reconstruction of the shape.

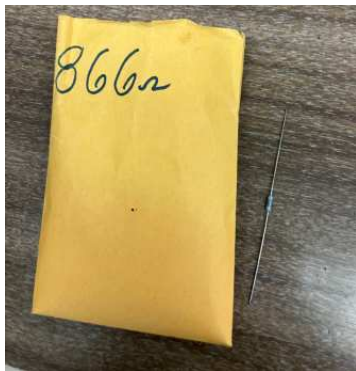
LABORATORY



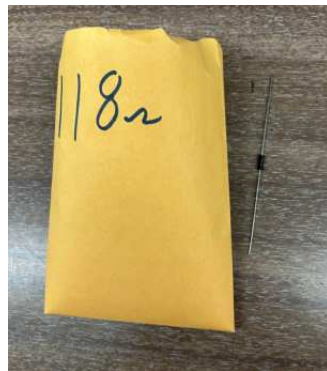
IMG 1. Sampling circuit, built in the lab. The exact circuit was used to get output on the spectrum analyzer, and a deviation of this circuit was used to observe PAM output on the oscilloscope.

1. Build the circuit of image 1. This is the sampling circuit. If you are using a LM348 op- amp, you should use a dual supply of $\pm 15\text{V}$. Keep in mind that in a quad op-amp, the power pins are used by all four op-amps, so that is why U1B is not powered separately above. The resistor R2 ensures that the source of the MOSFET and input of U1B are pulled to ground when the signal is not being passed. Use an analog input of sine at 10 kHz, 0.5Vpp, +0.5VDC offset. The offset is important to make sure your signal always stays positive.

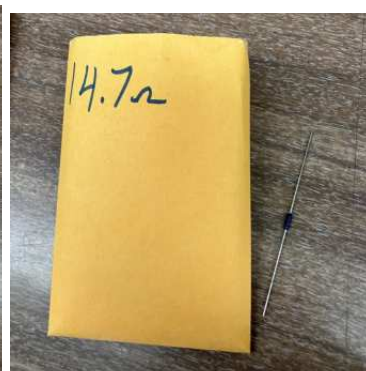
As a replacement for the 1k Ω resistor, 866 Ω , 118 Ω and 14.7 Ω resistors were used in series adding to a total of 998.7 Ω .



IMG 2. 866 ohm resistor



IMG 3. 118 ohm resistor



IMG 4. 14.7 ohm resistor

5 Pin Configuration and Functions

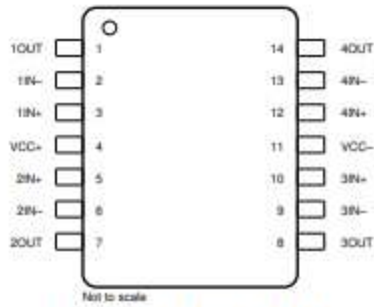
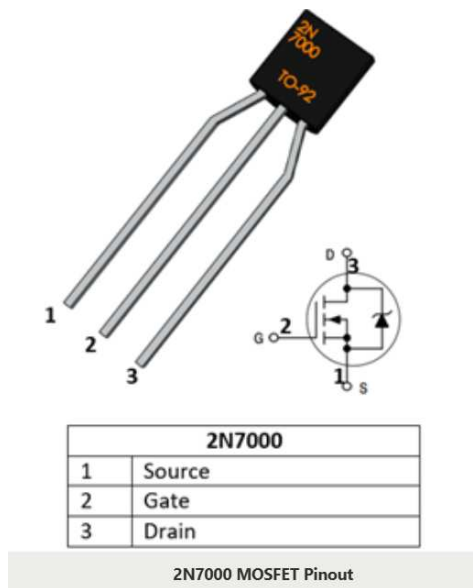


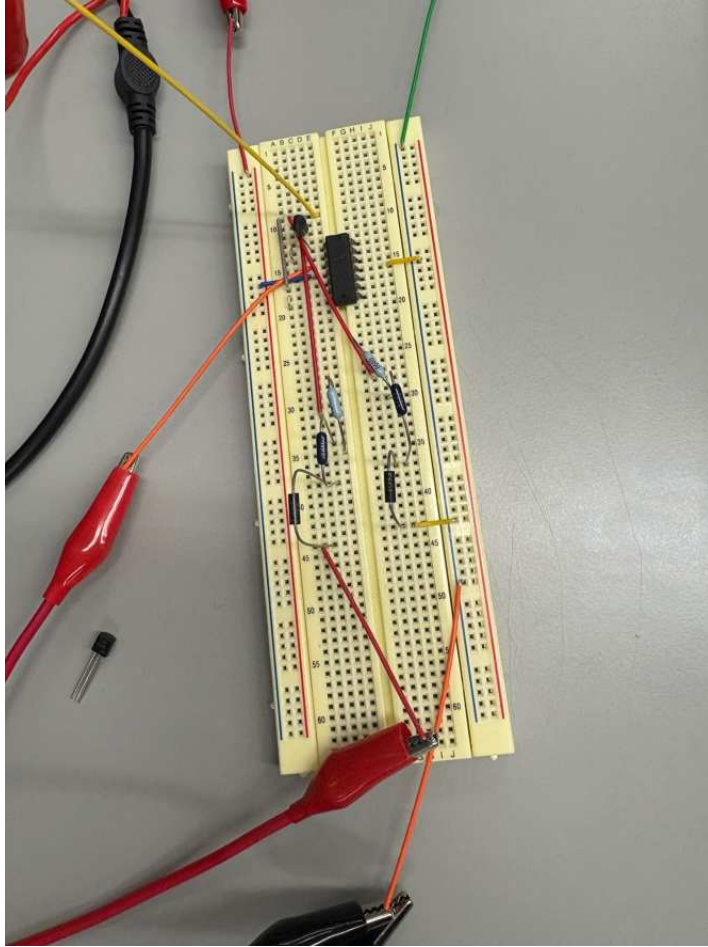
Figure 5-1. D, DB, J, N, NS, PW, W Packages,
14-Pin SOIC, SSOP, CDIP, PDIP, SO, TSSOP, CFP
(Top View)

IMG 5. Pinout of the LM 324AN

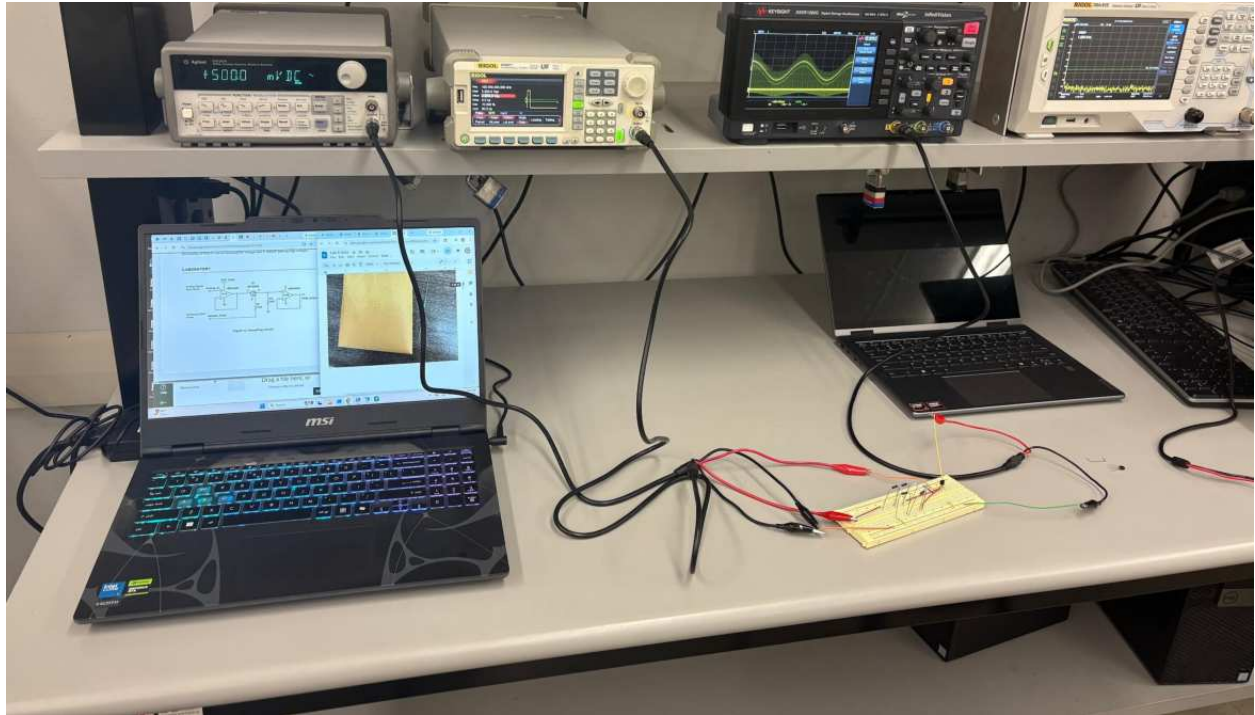
As far as the pinout and configuration of the LM 324An, a data sheet was utilized alongside the provided circuit schematic. The brand of the component and datasheet is Texas Instruments, found here: <https://www.alldatasheet.com/datasheet-pdf/pdf/2014700/TI/LM324AN.html>



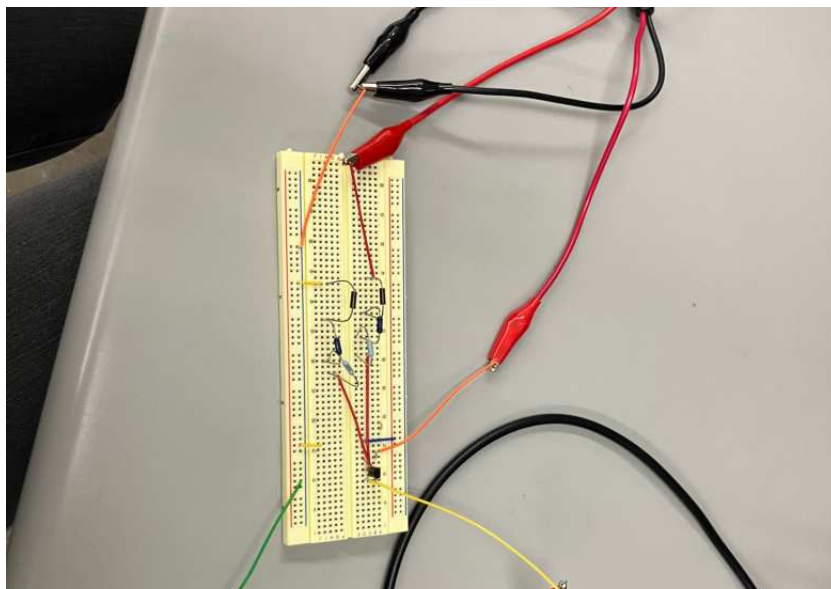
IMG 6. Pinout of the 2n7000G MOSFET, found at <https://components101.com/mosfets/2n7000-n-channel-mosfet>



IMG 7. *The circuit built based on image 1, for getting the spectrum analyzer output*



IMG 8. *set up of the circuit, using the below circuit customized for scope output as an example.*

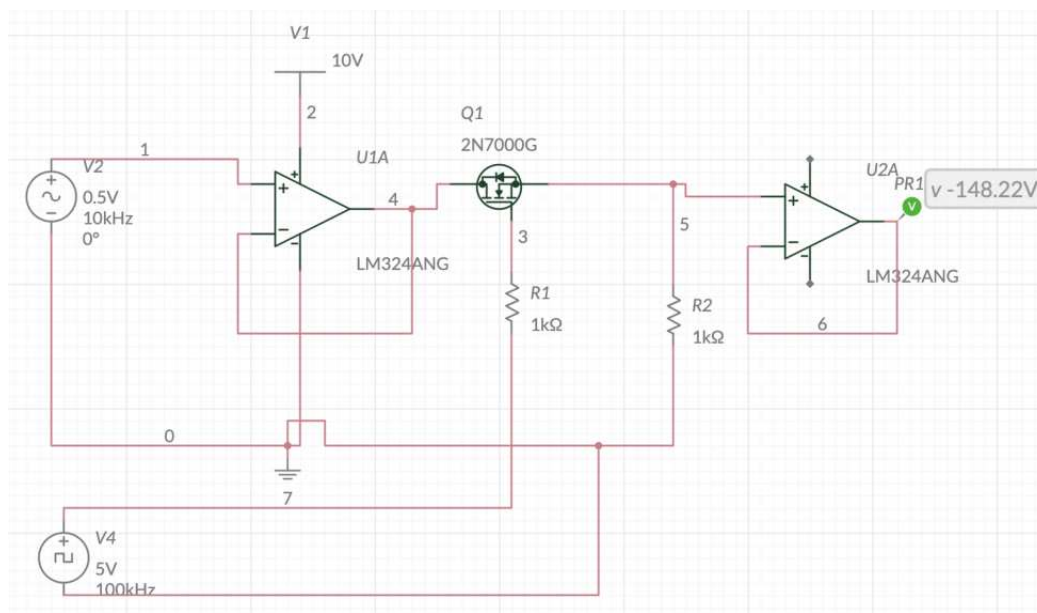


IMG 9. *The circuit built is a derivation of image 1, built to get output on the scope.*

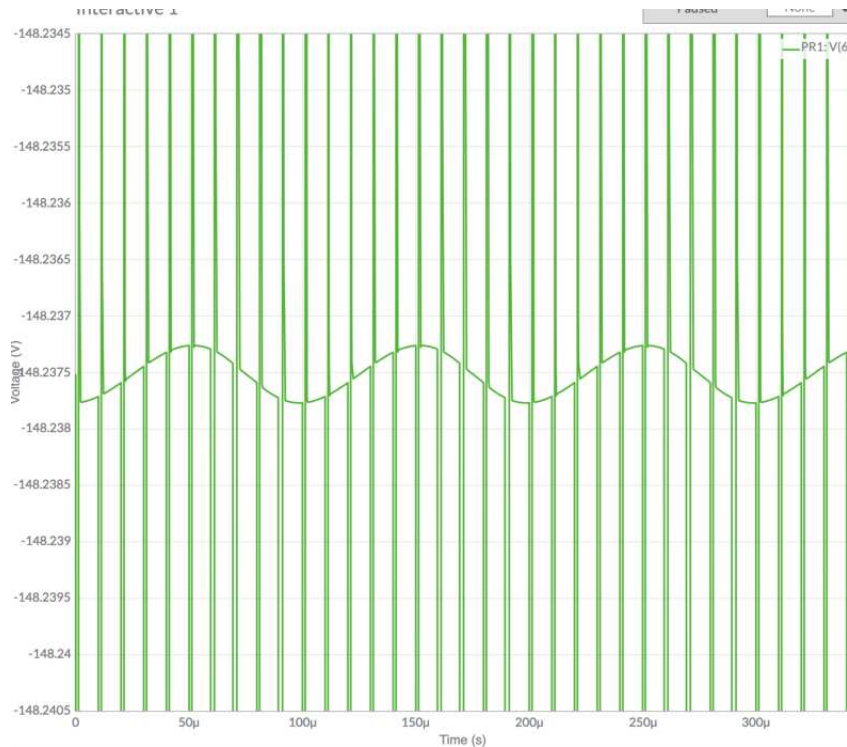
Removing the op-amp allowed the PAM signal to be observed more clearly. While building the sampling circuit the expected Pulse Amplitude Modulation (PAM) output was not clearly visible when using the LM324 op-amp. The main issue was the limited slew rate of the op-amp. The slew rate

determines how quickly the output voltage can change and in this case it was not fast enough to respond to the narrow sampling pulses. As a result, the op-amp output could not accurately follow the rapid switching, causing the PAM waveform to appear distorted and missing. To remedy this issue, the op-amp was removed from the circuit and the signal was observed directly after the switching element. After removing the op-amp, the output waveform more closely resembled the expected PAM signal, confirming that the op-amp was limiting performance. An alternative remedy would be to use an op-amp with a better slew rate, although it would be more expensive.

The main limitation in the circuit was the op-amp's inability to respond quickly enough to the high-frequency switching, but the transistor switching introduced additional noise. The MOSFET (transistor) acts as a switch that rapidly turns on and off to sample the input signal. However, because it switches very quickly, it introduces noise and distortion into the output. This switching action "chops" the signal, which is necessary for sampling but can make the waveform appear noisy on the oscilloscope. The high sampling frequency (100 kHz) compared to the input signal frequency (10 kHz) produces frequency components at the sum and difference of these frequencies. This results in spectral components at approximately 90 kHz and 110 kHz, and then multiples of such (ex: 190 kHz, 210 kHz and so forth). These frequency components are expected in PAM signals and are a result of the multiplication between the input signal and the sampling waveform.



IMG 11. *Circuit on Multisim*



IMG 12. Output of the Multisim circuit. Note that the output is not a perfect PAM wave due to the slew rate, as mentioned before

Analog input generator setting used for circuit 1, for spectrum analyzer output:



IMG 13. Analog input generator setting for spectrum analyzer output



IMG 14. Analog input generator setting for spectrum analyzer output



IMG 15. Analog input generator setting for spectrum analyzer output

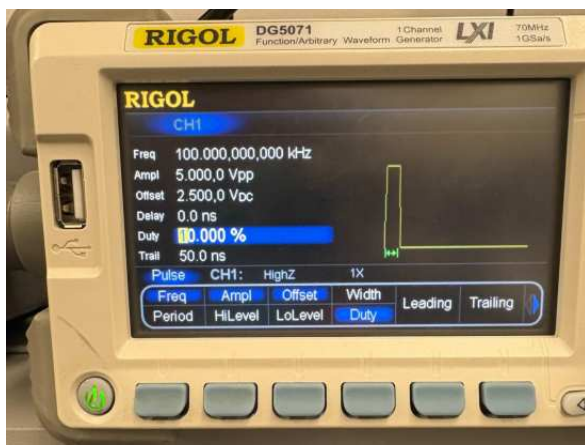
Analog input generator setting used for circuit 2, to get the output on the scope:



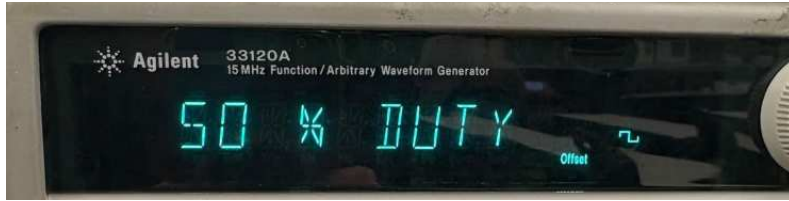
IMG 16. Analog input generator setting for scope output.

2. Set the clock generator to 100 kHz, 5Vpp, +2.5VDC offset with duty cycle at 10%. You might need to use the function called “pulse” when available to get a lower possible duty cycle. It is important to make sure our input signal is always positive, so don’t forget the step to add DC offset to the analog input signal.

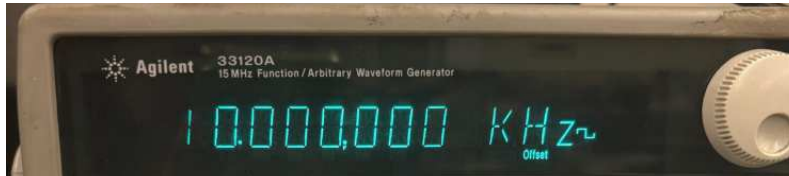
Clock output generator setting used for circuit 1 based off of image x and shown in image x, for spectrum analyzer output:



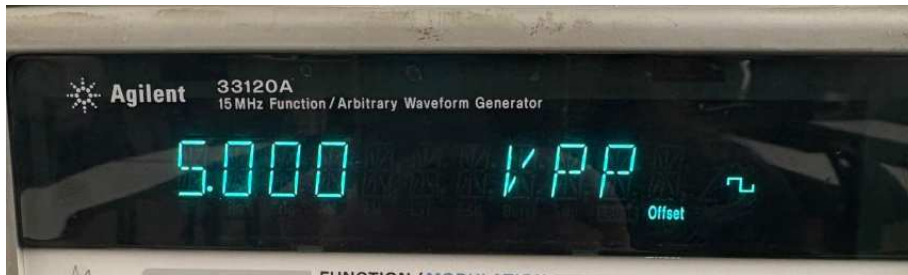
IMG 17. Clock generator setting for spectrum analyzer output



IMG 18. Clock generator setting for scope output



IMG 19. Clock generator setting for scope output



IMG 20. Clock generator setting for scope output

3. View the signal on both the oscilloscope and spectrum analyzer. Take care here to make sure your scales are setup to get a good screenshot on both instruments. This signal is difficult to capture using automatic triggering, so it would probably help to freeze the image or use single-shot triggering. For the spectrum analyzer, you can set these values:

Start freq = 0 Hz

Stop freq = 600 kHz

RBW (frequency resolution bandwidth) = 1 kHz

Then adjust the amplitude vertical controls (reference level and scale/div) to get a good picture. We are not concerned with any actual values of amplitude in this lab, just the picture.

4. If the Spectrum Analyzer is beeping, reduce your analog input signal (both VPP and DC offset), but make sure your signal remains always positive.

5. When satisfied the circuit is working as expected, capture screenshots, then also capture more screenshots at $f_s = 50$ kHz, 20 kHz, and 10 kHz. In other words, leave the analog

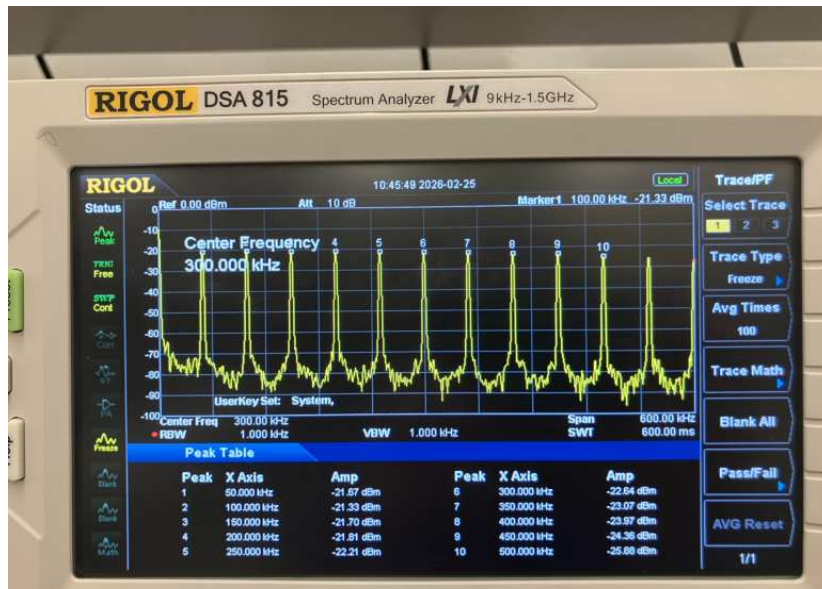
input at 10 kHz, and adjust the sampling frequency to the values above.



IMG 21. 10khz

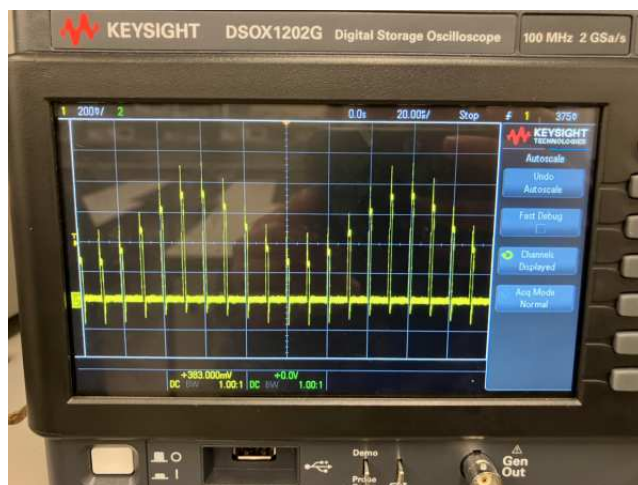


IMG 22. 20kHz

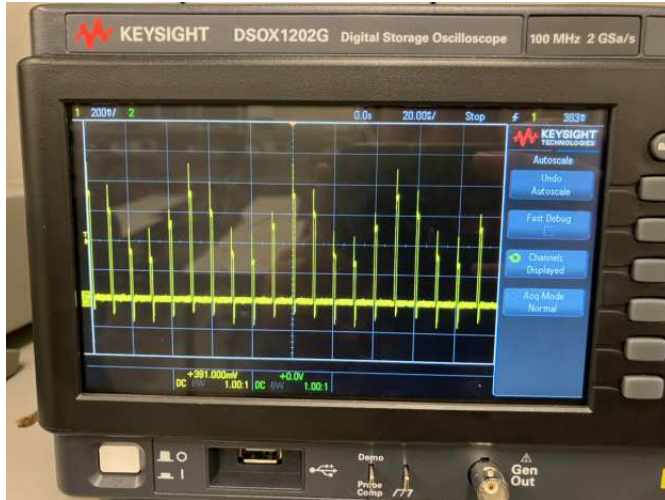


IMG 23. 50kHz

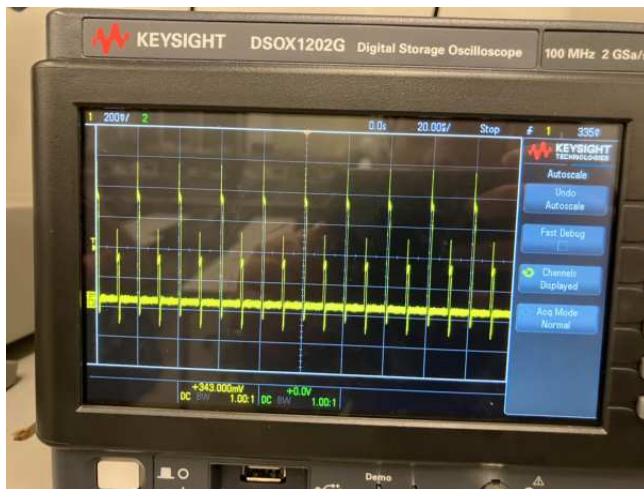
Using a new circuit for getting scope readings, we get a new output on the scope. This circuit does not have the chip and voltage source, but the Agilent function generator settings are the same. This new circuit was used to try to get around the limitations of the slew rate causing undesirable scope output.



IMG 24. 10khz



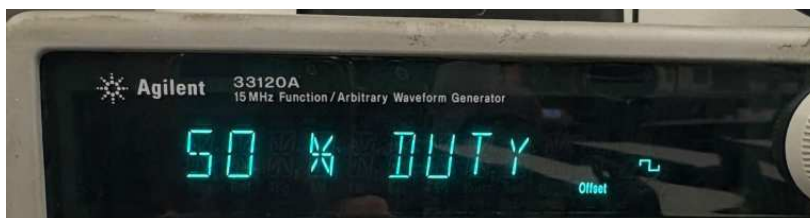
IMG 25. 20kHz



IMG 26. 50kHz

6. Switch your analog input signal (previously a sine wave) to 50% duty cycle square wave @ 10 kHz, and repeat screenshots with the sampling frequency varied as before to 100, 50, 20 and 10 kHz. Now we are sampling a square wave instead of a sine wave, so we should see the square shape in our sampled signal.

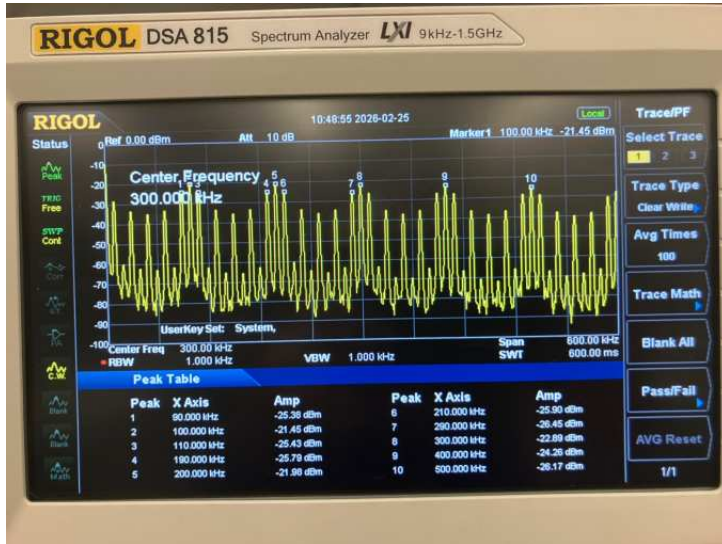
New settings for analog input signal for spectrum analyzer output:



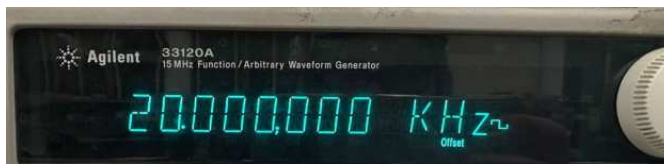
IMG 27. New settings for analog input signal for spectrum analyzer output circuit



IMG 28. 10kHz square wave



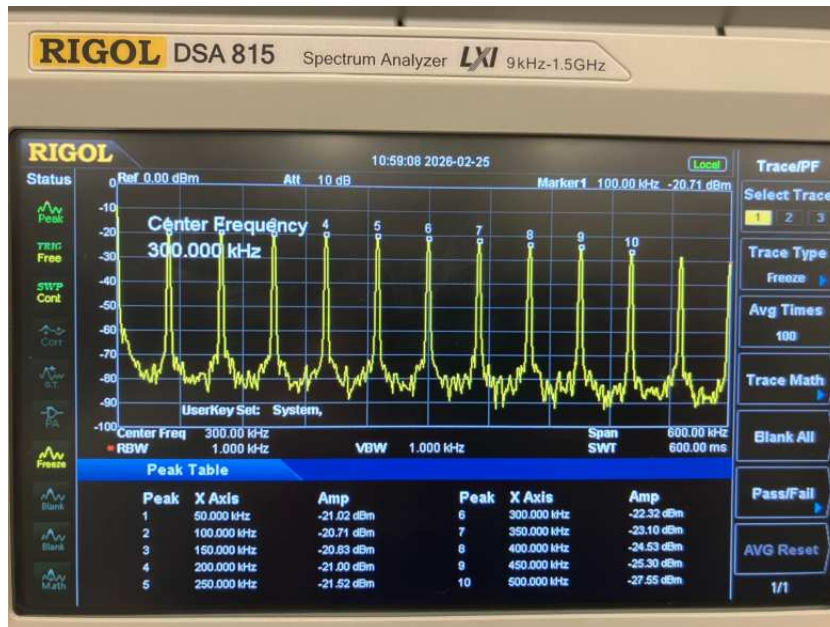
IMG 29. 10kHz



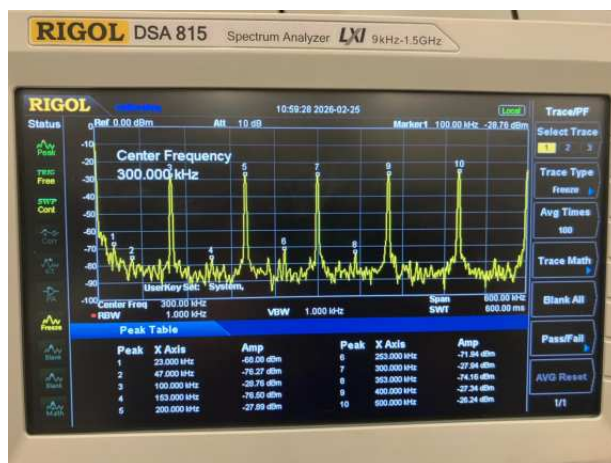
IMG 30. 20kHz square wave



IMG 31. 20kHz



IMG 32. 50khz



IMG 33. 100khz

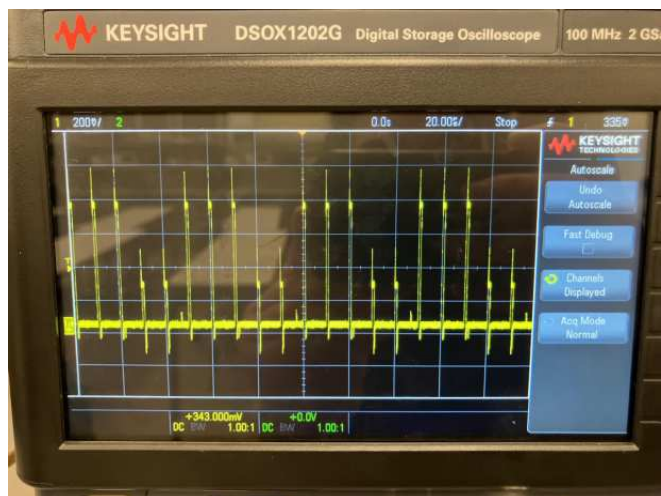
New settings for analog input signal for scope output:



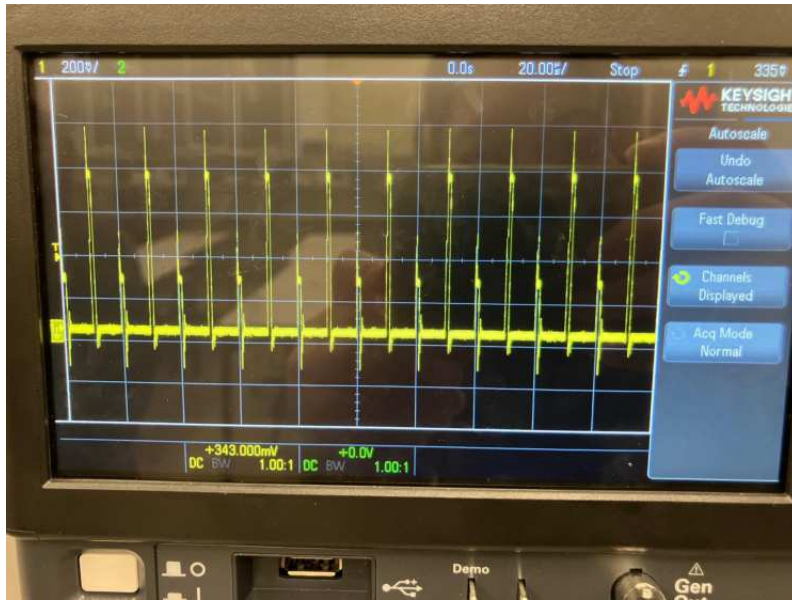
IMG 34. New settings for analog input signal for scope output circuit.



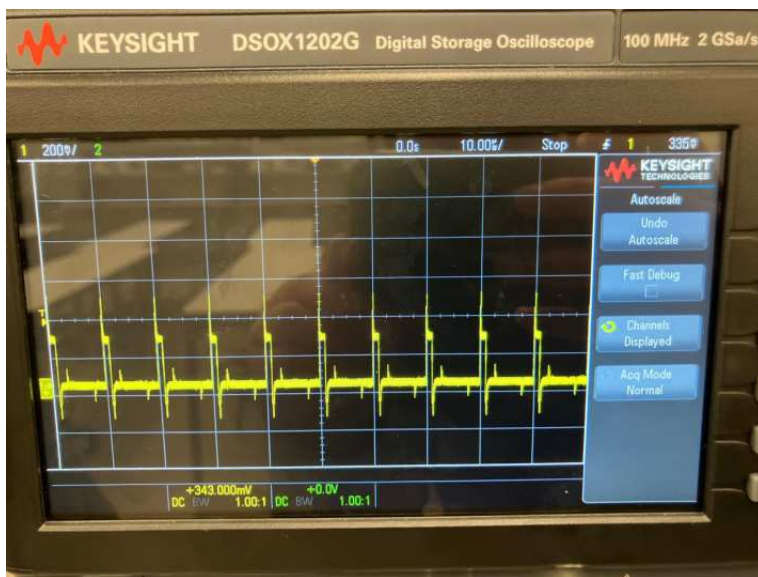
IMG 35. 10khz



IMG 36. 20khz



IMG 37. 50khz



IMG 38. 100khz

THEORETICAL ANALYSIS

1. Explain in your report the effect of sampling at the different frequencies you measured in the lab. Was the sine wave recognizable in the time domain when sampled at 10 kHz, 20 kHz, etc.?

When sampling at lower frequencies such as 10 kHz, the sine wave was not very clear in the time domain because there were not enough sample points to accurately represent the waveform. The signal appeared more jagged or distorted. As the sampling frequency increased to values like 20 kHz and higher

the sine wave became more recognizable because more points were captured per cycle. At higher sampling rates, the waveform looked smoother and closer to the original signal. This showed us that increasing the sampling frequency improves how accurately the signal is represented in time.

2. What is the ratio of the sampling frequency to the signal frequency for each measurement. Put this data into a table.

The ratio of the sampling frequency to the signal frequency shows how many samples are taken per cycle of the input signal. A higher ratio means better signal representation. For example if the signal frequency is 10 kHz and the sampling frequency is 10 kHz, the ratio is 1:1 which is not sufficient to properly represent the signal. If the sampling frequency is 20 kHz, the ratio becomes 2:1. Although this does meet the minimum requirement of the Nyquist theorem for accurately re constructing the frequency, we do not see a smooth waveform with an accurate shape yet. Once we get to 10:1 and above, we see a much clearer and more accurate representation of the signal.

Sampling frequency (kHz)	Signal frequency (kHz)	Ratio (fs/f)
10	10	1
20	10	2
50	10	5
100	10	10

3. At what sampling rate is it necessary to sample the message waveforms, according to the Sampling Theorem?

According to the Sampling Theorem (also known as the Nyquist Theorem) the sampling frequency must be at least twice the highest frequency present in the signal to accurately reconstruct the frequency and 10 times more to accurately represent the shape. This means the minimum sampling rate is $f_s \geq 2f$. However, while sampling at twice the frequency allows the signal to be reconstructed, it does not always produce a clear waveform. In practice, sampling at a higher rate that is even greater the 10 times the frequency gives a much better representation of the signal in the time domain.

4. Briefly explain the purpose of the two op-amps and the N-channel MOSFET.

The op-amps in the circuit are used to condition and process the signal. One op-amp is typically used to buffer the input signal so that it is strong and stable before sampling. The second op-amp helps

shape or restore the output signal after sampling. The N-channel MOSFET acts as a switch that rapidly turns on and off at the sampling frequency. When the MOSFET is on, it allows the signal to pass through, and when it is off it blocks the signal. This switching action creates the sampled signal, which is the basis of Pulse Amplitude Modulation (PAM). This was discussed in more detail in the introduction of this lab report.

5. Explain aliasing, how it occurs, and how can it be avoided.

Aliasing occurs when a signal is sampled at a frequency that is too low, causing the sampled signal to appear as a different frequency than the original. This happens when the sampling frequency is less than twice the signal frequency. To avoid aliasing, the sampling frequency must be at least twice the highest frequency of the signal as stated by the Nyquist Theorem.

Conclusion

In this lab we successfully examined the effects of sampling and implemented a Pulse Amplitude Modulation (PAM) circuit to observe both time-domain and frequency-domain behavior. By sampling a 10 kHz input signal at various sampling frequencies, we observed how the quality of the reconstructed signal depends heavily on the sampling rate. At lower sampling frequencies (such as 10 kHz), the waveform was poorly represented and appeared distorted due to insufficient sample points. As the sampling frequency increased (20 kHz, 50 kHz, and 100 kHz), the signal became progressively smoother and more recognizable confirming the importance of sampling above the Nyquist rate.

A major challenge encountered in this lab was related to the circuit design specifically the limitations of the LM324 op-amp. The op-amp's low slew rate prevented it from accurately following the rapid switching of the sampling signal, resulting in a distorted or unclear PAM waveform. This was seen in both the Multisim and the circuit we built on the breadboard. This issue was resolved by removing the op-amp and observing the signal directly after the MOSFET switching stage, which produced a waveform that more closely matched the expected PAM output. However, this introduced additional noise due to the fast switching behavior of the MOSFET, which shows that there's a trade off between signal clarity and meeting the limitations of the circuit.

Additionally, the lab reinforced the relationship between sampling frequency and using the spectrum analyzer. The spectrum analyzer showed frequency components at the sum and difference of the

input signal and sampling frequency (e.g., around 90 kHz and 110 kHz), which are characteristic of PAM signals. This demonstrated how sampling in the time domain affects the frequency domain representation.

Overall, this lab provided both theoretical and practical insight into sampling, PAM and aliasing. It emphasized the importance of proper sampling rates, effects of component limitations (such as the slew rate) and careful interpretation of real signals. The results aligned well with the contents of the laboratory hand out and expectations from the sampling theorem, confirming that higher sampling frequencies lead to more accurate signal representation and reduced distortion.